

OPERATING SYSTEMS

Process Management and Scheduling

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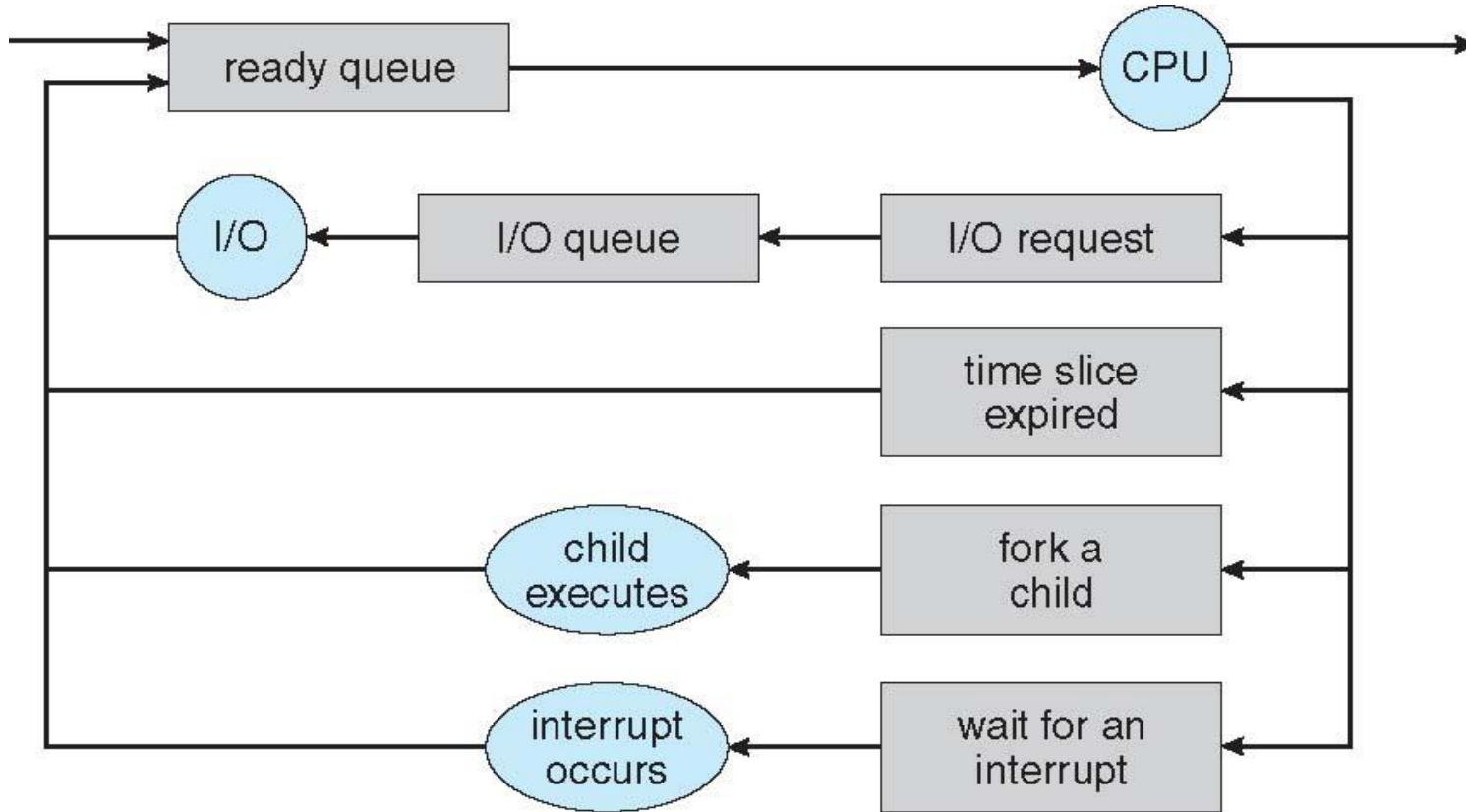
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- The objective of multiprogramming is to have some process running at all times, to maximize CPU utilization.
- The objective of time sharing is to switch the CPU among the processes so frequently that users can interact with each program while it is running.
- To meet these objectives the process scheduler selects an available process for execution on the CPU.
 - For a Single processor system, there will be never more than one running process
 - If there are more process, the CPU has to wait until the processesor is free and can be rescheduled.

- **Process scheduler** selects among available processes for next execution on CPU
 - Scheduling Queues, Schedulers, Context Switch
- Process Scheduling – Scheduling Queues – generally stored as linked list
- Process Maintains **scheduling queues** of processes
 - **Job queue** – set of all processes in the system
 - **Ready queue** – set of all processes residing in main memory, ready and waiting to execute
 - **Device queues** – set of processes waiting for an I/O device
- Processes migrate among the various queues

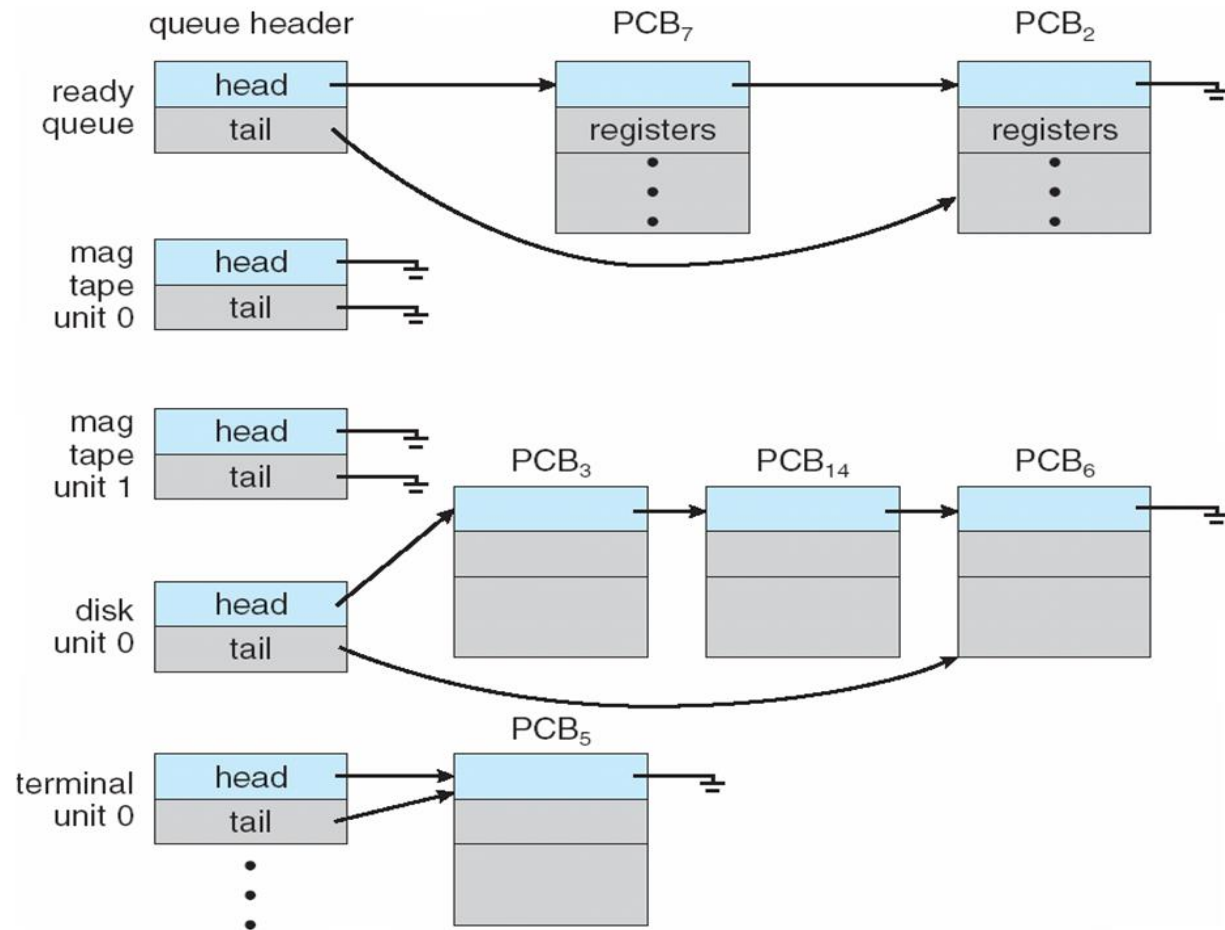
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Process Scheduling



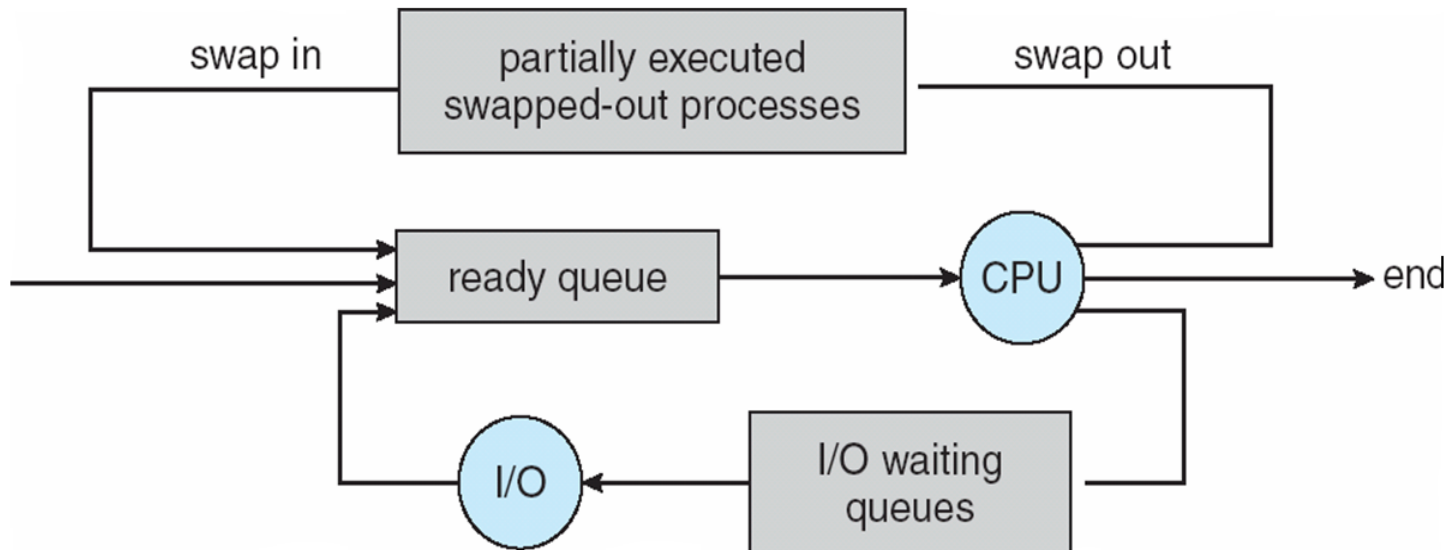
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Scheduling Queues



- **Long-term scheduler (or job scheduler)** – Selects process from secondary memory to main memory
- It happens infrequently. Process can happen slowly.
- It Controls the degree of multiprogramming.
- **Short-term scheduler (or CPU scheduler)** – Select processes from the ready queue and allocate the CPU to them. This happens frequently. Action should happen very fast.
- **Process are I/O bound or CPU bound**
 - **I/O-bound process** – spends more time doing I/O than computations, many short CPU bursts
 - **CPU-bound process** – spends more time doing computations; few very long CPU bursts

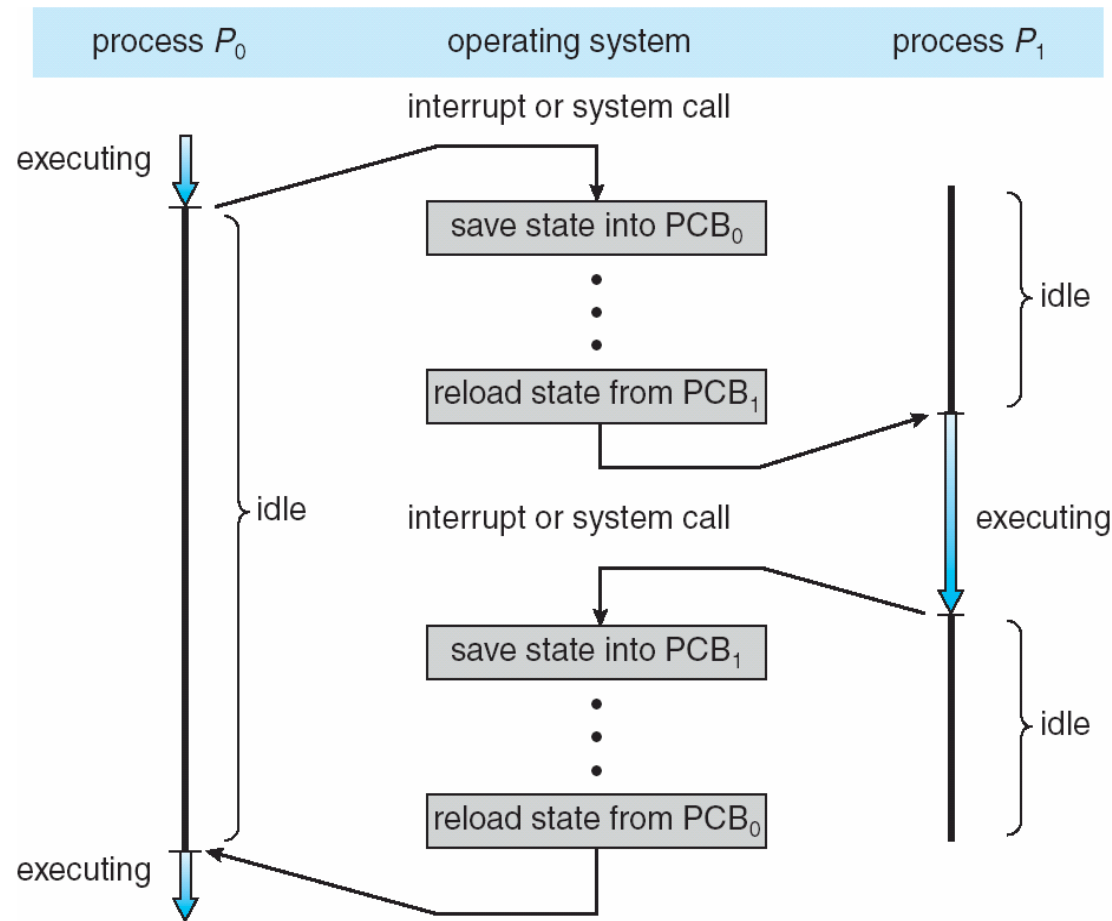
- **Medium-term scheduler** (or job scheduler) – removes process from memory so that reduce the degree of multiprogramming. Later, it reintroduces in main memory. i.e, it takes care of swapping process



- When CPU switches to another process, the system must save the state of the old process and load the saved state for the new process via a **context switch**.
- Save state of process0 and state restore of process1
- **Context** of a process represented in the PCB of the process
- Context-switch time is overhead; the system does no useful work while switching
- The more complex the OS and the PCB -> longer the context switch

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Context Switching





THANK YOU

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